

Quiz 1 — Practice Answer Key

STAT 184 — Summer 2026

Total: 30 points · 9 questions · 20 minutes

- Try the practice quiz first **without** looking at the key.
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1. (2 pts) Assignment

Answer: C (`score <- 100`)

Why: `<-` is the assignment operator we use in this class. `=` works in some places but causes confusion later. `==` is for *checking equality*, not assigning. `100 <- score` runs in the wrong direction. `assign()` exists but isn't taught.

2. (2 pts) Packages

Answer: B (`library(ggplot2)`)

Why: Installing is a **one-time** action per computer. Loading with `library()` happens **every session** — packages don't auto-load when R starts.

3. (4 pts, 1 pt each) Predict the output

(a) `sqrt(144)` → 12

(b) `c(10, 20, 30) + 5` → 15 25 35 (or `[1] 15 25 35`)

Why: scalar arithmetic on a vector is applied **elementwise**.

(c) `15 %% 4` → 3

Why: `%%` is the **modulo** operator (the remainder after division.) $15 \div 4 = 3$ remainder 3.

(d) `mean(c(1, 3, 5, 7, 9))` $\rightarrow$ 5

Why: $(1+3+5+7+9) / 5 = 25 / 5 = 5$.

4. (4 pts, 1 pt each) Identify the parts

(a) Data table: **Movies**

Why: starts with a capital letter, no parentheses after it.

(b) Functions: **any two of** `filter`, `summarise`, `mean`

Why: each is followed by an open parenthesis.

(c) Variable: **year** or **rating**

Why: lowercase, used inside function arguments.

(d) Constant: **2010**

Why: a single numeric value.

5. (3 pts) Data structures

Answer: A

A one-dimensional collection of values, all of the same type.

Why each other option is wrong:

- B describes a **matrix**.
 - C describes a **data frame**.
 - D describes a **list**.
 - E describes a single value, not a vector. (Technically a length-1 vector exists, but A is the better description.)
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6. (3 pts, 1 pt each) Function arguments

- (a) Function name: **round**
- (b) Number of arguments: **2** (`3.14159` and `digits = 3`)
- (c) The first argument: **positional** (it has no name on the left side of =)

Why: in `round(3.14159, digits = 3)`, the first argument is matched by *position*. `round()` expects `x` first. The second is matched by *name* (`digits = 3`).

7. (3 pts) Storing results

Full credit (3 pts):

Both lines compute the mean (which is 4), but:

- `mean(c(2, 4, 6))` just **prints** the result to the console. It's not saved anywhere.
- `m <- mean(c(2, 4, 6))` **assigns** the result to an object called `m`, so you can use `m` later.

Why: without `<-`, the value is computed and then discarded. The assignment operator is what stores the result for future use.

8. (4 pts, 1 pt each) Histograms

- (a) X-axis variable: **bill_length_mm**
- (b) Bin width: **2** (each bar covers 2 mm of bill length)
- (c) Inside color: **steelblue**
- (d) Outline color: **white**

Why `fill` vs. `color`: in `ggplot2`, `fill` is the inside of a shape and `color` is the border/outline.

9. (5 pts) Debug the code

- (a) **(2 pts)** The code uses `%>%` (or `|>`) between the `ggplot()` call and `geom_histogram()`, but `ggplot2` layers must be joined with `+`, not the pipe.

Why: the pipe `|>` (or `%>%`) is for **data verbs** like `dplyr` functions. `ggplot2` was built before the pipe became popular and uses `+` to add plot layers.

- (b) **(3 pts)** Correct version:

```
ggplot(penguins, aes(x = body_mass_g)) +  
  geom_histogram()
```

i Reminder

The real Quiz 1 will be **closed-book, closed-electronics, no AI**, and worth **3% of your final grade**. The questions will be different but cover the same topics.